

SLIDE 1

Problem & why it matters

Team name: <Team name>

Problem statement: **C1 FlatFair: HDB resale market intelligence & affordability forecasting**

The problem in one sentence:

HDB resale prices rose **56% since 2017** while household incomes rose **33%**. First-time buyers have no simple way to tell if an asking price, or the repayment that comes with it, is reasonable.

Who is affected, and how badly (use a number from open data):

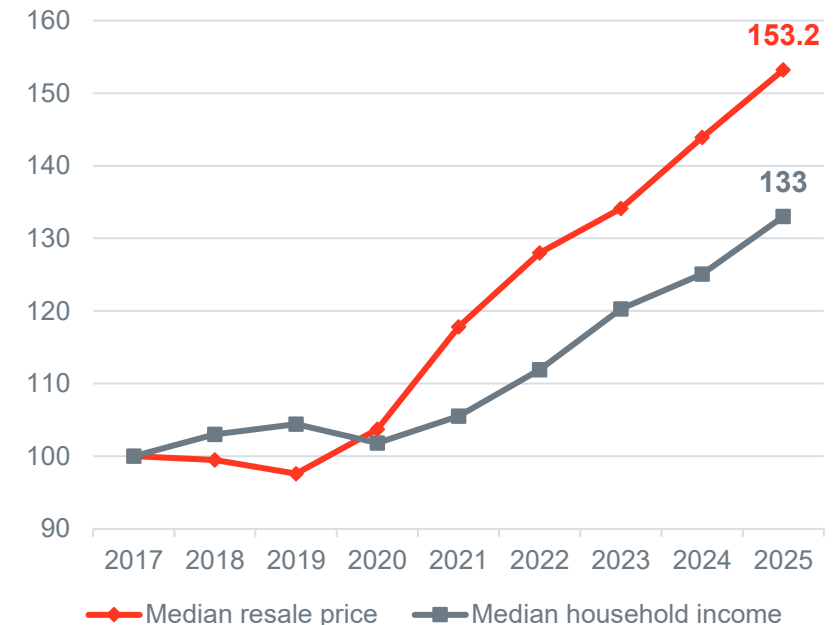
24,661 households bought a resale flat in the last 12 months. **1,847** of those flats sold for \$1 million or more. A median 4-room flat now costs **4.4 years** of the median household's income, up from 3.8 years in 2017.

Why it matters now:

Prices climbed 50% from 2020 and have now levelled off (**+1.3% in the past year**). A rising market no longer makes up for overpaying. Since August 2024 an HDB loan also needs a **25% downpayment**, up from 20%.

Prices have outrun incomes

Index, 2017 = 100



Annual medians. Sources: HDB resale flat prices (data.gov.sg); SingStat M810361 median monthly household employment income.

SLIDE 2

Solution & data

Your solution in one sentence:

FlatFair takes a buyer from “**what is the market doing?**” to “**is this flat, at this price, right for us?**” in five steps, using HDB’s own resale records.

Who uses it, and what decision it changes:

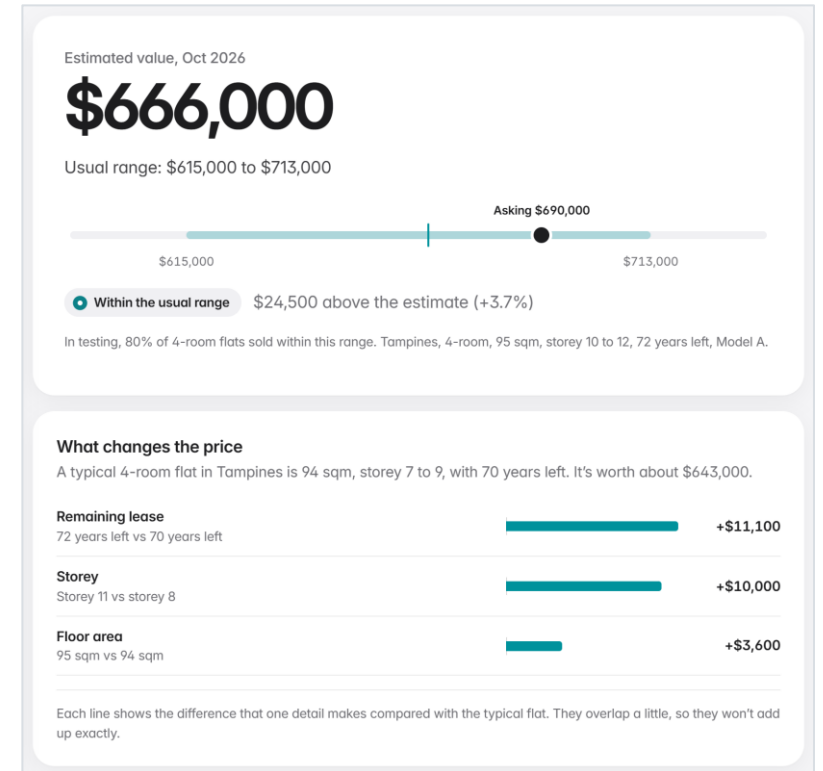
First-time resale buyers, and the housing counsellors and planners who advise them. It helps them decide **where to look** (towns that fit their budget), **what to offer** (asking price against the estimate and the 5 closest sales) and **when to buy** (a six-month forecast).

Datasets you will use (name + source, e.g. data.gov.sg):

1. **HDB Resale Flat Prices, Jan 2017 to present**, data.gov.sg (d_8b84c4ee...6abc), 241,920 sales
2. **Household Employment Income, key indicators (M810361)**, SingStat Table Builder
3. **Master Plan 2019 Planning Area Boundary**, URA via data.gov.sg (town map). Plus HDB Annual Report 2025 for supply context

What makes your approach different from a generic dashboard:

It prices one specific flat and shows the monthly repayment against the 30% limit. The price model was tested on **13,588 sales it had never seen**: typical miss **3.9%**, against 9.5% for the price-per-sqm rule of thumb. Every model has to beat a simple baseline. A 3-month average forecast better than our ML model, so we use the average.

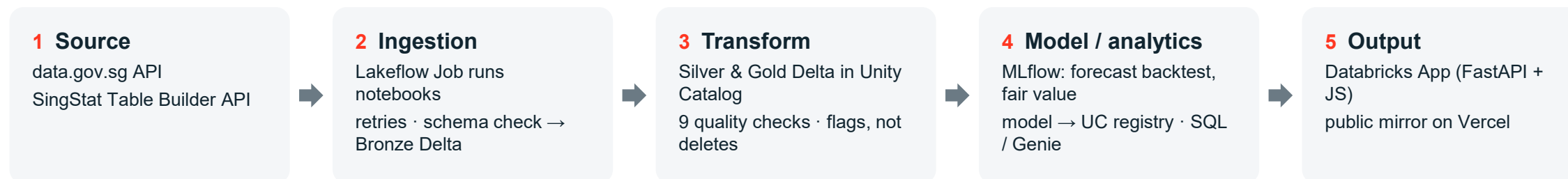


Working prototype: a Tampines 4-room asking \$690,000 sits within the usual \$615k to \$713k range. flatfair-nine.vercel.app

SLIDE 3

Databricks architecture & impact

Pipeline: source → ingestion → transform → model / analytics → output



(Name the Databricks components) **Lakeflow Jobs · Delta Lake · Unity Catalog (lineage, tags, model registry) · MLflow · Databricks SQL / AI-BI · Genie · Databricks Apps**

The user-facing output you will demo:

The FlatFair app: pick a town on a map of Singapore, then see its market, a six-month forecast, what you can afford, a fair price with comparable sales, and other towns side by side. **Already working end to end.**

Measurable impact if it worked:

On a typical \$630k flat, the usual pricing miss drops from **about \$60k to about \$25k**.
Buyers see the repayment against the 30% limit before they commit. The data refreshes every month.

What you can realistically finish in the two-week sprint:

Pipeline, both models, the app and 77 tests are built. In the sprint: run it as a Lakeflow Job in the final workspace, deploy on Databricks Apps, add a Genie space and dashboard, and test with five first-time buyers.